Nicholas Kantack

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Education

University of Oklahoma
Bachelor of Science: Engineering Physics
Graduated May 2016
Cumulative GPA: 3.98

University of Virginia
Master of Science: Electrical Engineering
Graduated December 2020
Cumulative GPA: 3.87

Work Experience

Amazon Web Services — July 2022 - Present
Software Development Engineer (L5)

- Backend development for Amazon Connect, a cloud-based customer service platform
- Principal architect for the rule-based and deep-learning-based algorithms powering Amazon Connect's estimated wait time feature
- Developed anomaly detection algorithms, automation tools for DevOps, and led effort to migrate legacy infrastructure to AWS Fargate

Johns Hopkins Applied Physics Laboratory — July 2018 - July 2022
Software Developer/Electrical Engineer

- Principal investigator for six internal research grants totaling $315,000 in funding
- Projects include AI in medicine, XR sensor fusion, state estimation for localization, and thin film RF device design

IntriCon Corporation — May 2016 - July 2018
Electronics Manufacturing Engineer

- Design and prototype sub-millimeter scale inductors for implantable medical devices
- Physical simulation and data analytics software design and support

Skills

Proficient AWS (ECS, Lambda, Fargate, EC2, Bedrock, IAM, SageMaker, DynamoDB, Connect), Java, Javascript, Python, PyTorch, Docker, infrastructure as code, probability theory & continuous statistics, statistical tests & data analytics, modeling & simulation, calculus & differential equations, machine learning, neural networks, Android, Tampermonkey, OpenSCAD, Agile, 3D printing, digital circuits, VR/AR

Familiar C++, C#, Tensorflow, CUDA, R, AutoCAD, Unity, Matlab, LaTeX, NodeJS, AngularJS, SQL, PCB design, RF systems, signal processing, embedded devices, NLP (HuggingFace tranformers), Deep RL, robotics, AWS Braket, quantum circuits

Publications and Projects

Visit nickkantack.com for more information.